

Modèle d'écoulement des Dépôts à Vue (DAV) — Refined Project Plan

Deliverable: a behavioral run-off curve $S(t)$ for non-maturing demand deposits (DAV), plus its rate/inflation **sensibilité**, packaged for IRRBB (and optionally FTP). **Data:** client-level (account-level) monthly panel, **2015-01** → **2024-12 (120 months)**, DZD. **External joins available:** market rate(s) + inflation (monthly).

This plan ports the rigor of the multi-horizon forecasting engine (CPCV, purge/embargo, sequential/block bootstrap, PIT calibration, no-look-ahead discipline, honesty-over-pretty-numbers) to a **survival / discrete-time-hazard** problem on a **short monthly panel**. Where the methods transfer directly they are reused; where the problem differs they are deliberately adapted, and the adaptation is justified rather than assumed.

0. The one thing that changes everything: this is forward extrapolation on 120 months

The trading engine estimated a *cross-sectional* signal and could afford CPCV as the **primary** protocol (combinatorial folds, 28 paths) because the quantity of interest (IC) is roughly exchangeable across folds. A run-off model is different on two axes:

1. **It extrapolates forward in time.** Its job is to predict the decay of *today's* stock over the coming months/years. A protocol that trains on the future to score the past (the core of CPCV) does **not** establish forward predictive validity — which is precisely what model validation and the regulator will demand. So **out-of-time (walk-forward) is the headline protocol; CPCV is a secondary variance/robustness tool**, not the deliverable's evidence base. This inversion is intentional.
2. **The binding uncertainty is temporal, not cross-sectional.** There may be 10^4 – 10^6 accounts but only **~10 years and very few rate/inflation cycles**. Cross-sectional sampling error is tiny; regime/macro uncertainty dominates. This is the same lesson as `feedback_report_ci_not_pvalues` (cross-asset-honest block bootstrap, block = horizon): **accounts are not independent — they share macro shocks** — so an account-only bootstrap understates total uncertainty exactly the way the per-asset-independent bootstrap did. The honest headline interval must come from a **time-block / common-factor bootstrap**, not an account bootstrap. See §6.

Everything below follows from these two facts.

1. Validation budget — how to split 120 months

1.1 Frozen out-of-time (OOS) test

Reserve the last 18–24 months as a frozen out-of-time test; default = 24 months (2023-01 → 2024-12). Touch it **once**, at Phase 5, for the realized-vs-predicted backtest. Development uses **2015-01 → 2022-12 (96 months)**.

Rationale: - 24 months ≈ 20% of the timeline, covers ≥2 full seasonal cycles, and lets you backtest run-off curves out to a 12–24-month horizon from rolling origins inside the window. - It is long enough that the OOT backtest is a genuine multi-month run-off test, not a one-shot.

Data-contingent caveat (decide with Gate A first). With Algerian rates historically administered, the *only* episode where rates/inflation actually moved may be narrow (e.g. the post-2022 inflation spike). There is then a real tension: - Put that episode in the **test** window → you test the rate channel out-of-sample (ideal) but cannot *fit* the elasticity well. - Put it in **development** → you can fit the elasticity but it is then OOS-untested.

Action: inventory *when* rate and inflation actually varied (Gate A) **before** fixing the split. Prefer a split that leaves at least one macro-varying episode in *both* dev and test. If there is only one such episode, keep it in dev (so the elasticity is estimable) and **declare the rate sensibilité as OOS-unvalidated** in the validation note (honesty rule). Do not silently choose.

Horizon-vs-history limit (state it openly). If the run-off horizons of regulatory interest exceed ~24 months, 10 years of data **cannot** fully validate them out-of-sample. Validate the hazard and the ≤24-month aggregate empirically; beyond that the curve is a **structural extrapolation** and must be labelled as such, with sensitivity to the extrapolation assumption reported.

1.2 Development window (96 months): two nested protocols

- **Primary — anchored walk-forward (expanding origin).** Train on $[2015-01, o]$, score $(o, o+K]$, step the origin o forward. This is the protocol that mirrors deployment and selects hyper-parameters and the model class. Report the **mean and dispersion across origins**, never a single fold.
- **Secondary — Combinatorial Purged CV (CPCV), adapted.** Used only to get a *distribution* of performance and a stability read, analogous to the 28-path engine. Group the 96 dev months into **8 super-blocks of ~12 months → N=8, k=2 → C(8,2)=28 paths**, with **purge + embargo** between train and test super-blocks (§1.3). Grouping unit = calendar month, so an account's rows are split by time and the *same month* never appears in both train and test. Report fold dispersion as a robustness check, **not** as the headline number.

1.3 Purge & embargo sizing

The single-month hazard label at (account i , month t) = event in $(t, t+1]$, so raw label overlap is **1 month**. But the deliverable aggregates hazards into a survival curve over a horizon H ($S(t) = \prod(1 - \hat{h})$ out to H), and the **backtest** of S over H months has overlap H . Therefore:

- **Embargo / purge = H**, the maximum run-off horizon you evaluate (e.g. 12 or 24 months), applied on **both** sides of every test block. Any train person-month whose $[t, t+H]$ label window overlaps a test block is purged. This is the direct analog of the engine's embargo $E = 60$ business days = max horizon.
- For pure hazard discrimination (1-month), a 1-month embargo suffices; for $S(t)$ calibration use H .

1.4 Optional: unseen-account generalization split

The time split lets an account appear in train (early months) and test (late months) — desired, it is the same entity over time (like an asset). If you also need to claim generalization to **newly opened accounts**, additionally hold out a random sample of accounts entirely and report a “new-client” score. Optional, secondary.

2. Phase 0 — Scope & data contract (1½ week)

Unchanged in intent; tightened on the points that bind the statistics.

- Pin with the manager: GenAI interpretation, segment/currency in scope, downstream consumer (IRRBB sensitivity / FTP / both), and the *référentiel* definition of the `CODE TYPE COMPTE` values (B35M/T cut). Confirm grain = **account-month**.
- **Define the event precisely** — closure / balance below a floor / dormant N months — and decide **competing risks vs single event** here (see §3.1). Everything downstream keys off this.
- **Define the run-off quantity precisely.** Is $S(t)$ the decay of the *current outstanding stock* (standard écoulement) driven by (a) account attrition **and** (b) balance erosion on surviving accounts? If yes, the model is not pure survival — it is $B_i(t) = B_i(0) \cdot A_i(t) \cdot r_i(t)$ (alive-indicator $\times$ balance-ratio-given-alive). Decide whether to model $r_i(t)$ explicitly or freeze balance at t_0 and absorb erosion into the hazard. **This choice must be made now**, not discovered in Phase 3.
- Fix success metric now (see §5): OOT realized-vs-predicted aggregate $S(t)$ + calibration + interval coverage, benchmarked against the convention and the ECM.
- **Output:** one-page scope note (incl. event definition, run-off quantity, frozen-OOS decision pending Gate A).

3. Phase 1 — Data engineering: build the panel (1–1.5 weeks)

Goal: a clean account-month panel with **survival bookkeeping** and **PIT-correct** macro joins.

- Parse fixed-width dumps (`read_fwf` / regex) $\rightarrow$ tidy DataFrame; one row per (`account`, `month`) with balance (`CTRLV KDA` for currency consistency), type, `DATE OUVERTURE`, currency.
- Filter to a homogeneous population (DZD, target segment).
- **Survival bookkeeping:** event flag; **right-censoring** for accounts alive at 2024-12; **left-truncation** for accounts opened before 2015 (they enter at 2015-01 with seasoning > 0 — the hazard likelihood must condition on survival-to-entry, do **not** treat entry as origin).
- **Features (all causal, $\leq t$):** seasoning (months since `DATE OUVERTURE`), age cohort, lagged balances, balance buckets, calendar month/year, intra-account balance trend/volatility.
- **Macro join — PIT discipline (direct port of the Step-3 look-ahead lesson).** Rates/inflation are published with a lag; align **as-of** the information set at decision month t , never the final/revised vintage aligned to its reference month. Document the assumed publication lag.
- **No-look-ahead on fitted statistics.** Balance-bucket thresholds, standardization scalers, ECM coefficients, frailty estimates, any quantile cut — **fit on the training window only**, applied forward. This is the exact failure mode of the engine’s global-fit bug (`feedback_step3_lookahead_bug`): a threshold fit on 2015–2024 and used at 2016 leaks the future. Make `fit_end` a **required** argument, not a default.
- **QA:** missingness, gaps, outliers, currency consistency; **reconcile summed CTRLV KDA to a known aggregate total** per month (the single most important data-integrity check).
- **Output:** modeling-ready panel + data-quality memo.

→ Gate A (rate identifiability) — run before any rate-conditional model

Check the market-rate series actually **varied** over 2015–2024 (plot levels, count distinct regimes, measure its standard deviation and number of moves). If rates were largely administered/flat, the rate sensitivity is **weakly identified** — flag it now, lean the stress story on **inflation** and on **scenario assumptions** rather than a fitted rate elasticity, and let this gate decide both the frozen-OOS placement (§1.1) and how strong the *sensibilité* claim may honestly be.

4. Phase 2 — Baselines: convention + aggregate ECM (1 week)

The floor to clear and the regulator-recognized reference.

- **(a) Convention baseline.** Core/volatile split via minimum-balance or moving-average floor, then assign amortization to each sleeve. This is the recognized reference; the model must **beat it on the OOT backtest** to justify itself.
- **(b) Aggregate ECM.** Collapse to aggregate (or per-segment) monthly balance; fit an error-correction regression on rate(s) + inflation + seasonality $\rightarrow$ equilibrium level, reversion speed, interpretable rate **elasticity**. This is the “classic OLS” floor and yields an aggregate run-off + sensitivity benchmark.
 - **Rigor add:** the ECM is a single time series of ~96 points with serial dependence — its coefficient CIs from textbook OLS are **wrong**. Use **HAC (Newey–West)** standard errors and a **moving-block bootstrap over months** for the elasticity CI. This is the small-sample, serially-dependent analog of the engine’s HAC/ `n_eff` treatment.
- **Output:** baseline $S(t)$ curves + elasticity number with honest (block-bootstrap) intervals.

5. Phase 3 — Core model: account-level discrete-time hazard (2–3 weeks)

The behavioral run-off and the project backbone.

5.1 Structure

- Reshape to the **person-month** survival table: (account, month, event_flag, features, balance_weight), respecting left-truncation (rows only from entry month onward) and right-censoring.
- Fit a **discrete-time hazard**: start with **pooled logistic** (interpretable, the documented baseline), then **gradient boosting** as the challenger that typically wins on tabular-temporal panels.
- Better-method (v2) inline — competing risks**. If closure / dormancy / large-withdrawal differ in balance impact (decided in Phase 0), fit a **multinomial / cause-specific discrete hazard** rather than a single binary event; aggregate cause-specific survival into the stock decay. Generic survival regression that lumps causes will mis-state run-off if the causes have different balance signatures.
- Better-method (v2) inline — monotonic constraints**. Constrain the GBM to be monotone in seasoning where theory demands it (e.g. hazard non-increasing in tenure for sticky deposits), both for governance defensibility and to reduce variance on the short panel.

5.2 Weighting, imbalance, calibration

- Balance-weight the loss** (CTRLV L KDA): the aggregate is whale-dominated, so the economically relevant loss weights each person-month by its balance. This is the run-off analog of the engine's uniqueness weights — same machinery (sample weights into the loss), different (economic) motivation. Optionally combine with inverse-censoring weights (IPCW) for unbiased survival estimation.
- Handle rare-event imbalance (low monthly hazard) via the weighting and threshold-free metrics; do **not** resample in a way that breaks the time structure.
- Calibrate the hazards** — reliability diagram + **PIT** check (the buy-side calibration instinct ports directly; cf. project_pit_calibration_2026_06_11). A run-off model lives or dies on calibration, not discrimination: a well-calibrated mediocre-AUC hazard gives a correct $S(t)$; a sharp-but-miscalibrated one does not.

5.3 Validation (strict, per §1)

- Primary**: anchored walk-forward; **secondary**: 28-path CPCV with embargo = H . No leakage: fitted statistics fold-local, macro PIT-aligned.
- Aggregate per-account survival $\times$ balance $\rightarrow$ portfolio $S(t) = \sum_i S_i(t) \cdot B_i(t) / \sum_i B_i(0)$ (with the balance-erosion term $r_i(t)$ if modeled per Phase 0).
- Feed rate/inflation scenarios through each $S_i \rightarrow$ **stressed $S(t)$** .
- Report intervals, not point curves** — and the *right* intervals (§6).
- SHAP** for interpretability and the model-validation file.
- Output**: base + stressed run-off with honest intervals + SHAP attributions.

→ Gate B (is deep learning worth it?)

If Phase 3 already beats the convention and validates cleanly out-of-time, the deep/GenAI layer is a genuine **option**, not a requirement. Decide here whether Phase 4 buys enough lift to justify its governance cost.

6. Uncertainty quantification — the honest-CI core (applies across Phases 2–5)

Two independent error sources; **report both, headline the binding one**.

- (a) Cross-sectional (account) uncertainty** → **account-cluster bootstrap**. Resample *accounts* with replacement, carrying **all** of each account's month-rows (block = the account), recompute $S(t)$. With many accounts this interval is **narrow**.
- (b) Temporal / common-shock uncertainty** → **stationary/moving block bootstrap over months**. Block length $\approx$ macro autocorrelation length (or $= H$); resample contiguous month-blocks and refit. With ~ 10 years and few cycles this interval is **wide** and is the **true binding constraint**.

The honesty rule (direct port of feedback_report_ci_not_pvalues): accounts share macro shocks, so they are **not** independent. An account-only bootstrap understates total uncertainty exactly as the per-asset-independent bootstrap did (it turned a true $p=0.34$ into a spurious $p<0.001$). The **headline fan-chart and any p-value must come from the time-block bootstrap** (or a hierarchical bootstrap: resample months $\rightarrow$ then accounts within), not the account bootstrap. Show the account bootstrap only as the (small) idiosyncratic component, clearly labelled.

Model-side reinforcement — frailty / latent common factor. Add a shared random effect (account frailty) or a **latent monthly factor** so the model itself attributes correlated attrition to a common shock. Without it, scenario fan-charts are too tight (idiosyncratic-only) — the same under-dispersion failure seen in PIT calibration. The latent factor also makes the stressed- $S(t)$ fan honest.

6.1 Conformal prediction for distribution-free stress intervals

The block bootstrap gives **parameter/sampling** uncertainty; conformal gives **distribution-free finite-sample coverage** on the prediction itself — and the two are complementary, not substitutes. Port the engine's conformal layer (Part VI / Ch37, project_pit_calibration):

- Object to conformalize**. Define a nonconformity score per evaluated unit — per (segment, horizon) realized-vs-predicted run-off residual $|S_{\text{real}}(t) - \hat{S}(t)|$ (or the per-month hazard residual). Calibrate on the most recent dev block; apply forward.
- Split conformal** gives marginal coverage **if exchangeable** — which time series violate. So use **weighted conformal** (Tibshirani et al., reweight calibration points toward the test regime) and/or **adaptive conformal (ACI / Gibbs–Candès)** that updates the quantile online as coverage drifts. This is the run-off analog of “weighted conformal under TS violation + per-CPCV-path quantile averaging.”
- CQR** (conformalized quantile regression) sits naturally on a quantile/survival head: fit quantile hazards, then conformalize the residual $\rightarrow$ bands with coverage guarantee that *adapt* to local difficulty (wider in stressed months).
- For stress specifically**: wrap each stressed $S(t)$ scenario in a conformal band so the IRRBB deliverable carries a **coverage-guaranteed** interval, then **widen by the time-block bootstrap** for macro-parameter uncertainty. Report the *combined* band. The headline stress claim is then: “under a +200bp shock, run-off at 12m is $\hat{S} \pm [\text{conformal} \oplus \text{block-bootstrap}]$ with $\sim 90\%$ coverage.”
- Honest caveat to state**: conformal coverage is only as exchangeable-valid as the calibration regime resembles the stress regime. Under a genuine out-of-distribution rate shock (which Algeria may never have realized — Gate A), conformal coverage is **not guaranteed**; the

band is then a *scenario* band, not a coverage-certified one. Say which it is.

7. Phase 4 — GenAI / deep challenger (optional, 2–4 weeks)

Kept only if it beats Phase 3 on the **same** OOT protocol. Two routes, **not both**:

- **(A) Per-account temporal model.** One **global** shared TCN or temporal transformer with account as the batch dimension (the engine’s “global model, asset = batch” pattern). **Two-clock encoding**: calendar positional encoding + age/seasoning positional encoding (the engine’s calendar+temporal embeddings, adapted). NaN masking (masks-not-drop). A proper **discrete-time survival head** (DeepHit-style cause-specific or a hazard logit per month), **balance-weighted**. Optionally **depth-2/3 signature features** of each account’s (balance, rate, inflation) path as inputs (ports Part II/Ch22 directly), and the **latent monthly factor** from §6 for residual common shocks.
 - v2 note: prefer a *proper survival head + monotone/competing-risk structure* over a generic regression head — a behavioral run-off needs calibrated hazards, not just point forecasts.
- **(B) Time-series foundation model, zero-shot** on the aggregate/segment balance series as the literal “GenAI” garnish, benchmarked vs the ECM. Honest framing: this is a benchmark/garnish, not the engine.

Benchmark honestly against Phases 2–3 on identical OOT splits, same embargo, same balance-weighted metric. **Deflate** for the number of architectures/event-definitions tried (the engine’s `eval_path_0` cherry-pick and deflated-Sharpe lessons: don’t report the best of N as if it were the only one). **Output**: challenger results + keep/drop recommendation.

8. Phase 5 — Sensitivity, stress & validation packaging (1–1.5 weeks)

- **Sensibilité.** Re-run $S(t)$ under rate shocks ($\pm 200\text{bp}$ parallel, steepener/flattener) and inflation scenarios $\rightarrow$ effective **duration / key-rate sensitivity / WAL** of the deposit. Carry the Gate-A caveat through: if rate variation was weak, present rate sensitivity as **scenario-driven, not empirically identified**, and lead with inflation.
- **Robustness battery (ports the engine’s model-free safeguards):**
 - stability of $S(t)$ across time windows and across segments;
 - sensitivity to the **event definition** (re-estimate under the alternative closure/dormancy/floor definitions — this is a known degree of freedom, so quantify it);
 - **realized-vs-predicted** aggregate $S(t)$ on the frozen OOT window, with **interval coverage** (does the 90% time-block fan cover realized?) as the headline calibration metric;
 - convention-vs-ECM-vs-model on identical OOT origins.
- **Metrics summary.** Discrimination (time-dependent AUC / Uno’s C, IPCW) is secondary; **calibration (reliability + PIT) and aggregate- $S(t)$ error + coverage are primary**, plus predicted-vs-realized WAL/duration.
- **Model-validation file.** Assumptions, data lineage, PIT/no-look-ahead evidence, baseline-vs- challenger table, calibration evidence, SHAP interpretability, and **honest limitations** — especially the Gate-A rate identifiability and the horizon-vs-history extrapolation limit.
- **Output**: final run-off curves (base + stressed), the run-off **rule**, the sensitivity table, and the validation note.

8b. Explainability — model *and* data (governance-grade)

A model-validation / IRRBB context demands you can explain **both what the data says** and **why the model decides**. Build explainability in three tiers, anchored on interpretable baselines so the “AI” never has to be taken on faith.

Tier 1 — interpretable anchors (always present). - The **ECM elasticity, equilibrium level, reversion speed** are inherently readable economic numbers — they are the explanation the manager already named. - The **pooled-logistic hazard** gives signed, magnitudable coefficients (seasoning, balance bucket, calendar) as a transparent reference the challenger is benchmarked against. - **Monotonic constraints** on the GBM (hazard monotone in seasoning, etc.) make the model’s behavior explainable *by construction* and defensible to validation.

Tier 2 — data & feature explanation (model-agnostic). - **SHAP / TreeSHAP** global (mean $|\phi|$) for what drives attrition, and **local SHAP** for “why this cohort/segment runs off faster.” Compute **out-of-bag/out-of-time** to avoid the in-bag optimism bias (Ch41 lesson). - **Partial-dependence / ALE curves** for seasoning, balance bucket, rate, inflation — these *are* the behavioral story (e.g. “hazard drops sharply after the first 6 months, then flattens”). - **Balance-weighted importance**: report importance weighted by `CTRL KDA`, because what explains the *aggregate* run-off (whale-dominated) differs from what explains a random account. State both.

Tier 3 — deep-model introspection (only if Phase 4 is kept). - For the **temporal transformer, attention maps** show *which months and which clock* the model attends to — the **two-clock** design lets you separate **calendar attention (seasonality)** from **age/seasoning attention (tenure effects)**, which is a genuinely useful behavioral read. - **Caveat (state it in the validation note)**: attention weights are **not** a faithful explanation on their own — high attention $\neq$ high causal contribution. Corroborate any attention story with **SHAP on the deep model, ablation** (remove the rate channel, measure $S(t)$ shift), and **integrated gradients / occlusion** before claiming a mechanism. Attention is a hypothesis generator, not the evidence. - For the **latent monthly factor** (§6), plot its trajectory against macro — if it tracks the rate/ inflation episode, that *is* the explainable common-shock channel; if it doesn’t, it’s absorbing unexplained correlated attrition, which you must disclose.

Deliverable: a short interpretability appendix in the validation note — Tier 1 numbers, Tier 2 SHAP/PDP plots, and (if applicable) Tier 3 attention + ablation, each labelled by how much evidential weight it carries.

9. The spine, in one breath

Build the PIT-correct account-month panel $\rightarrow$ **check rates actually moved (Gate A)** $\rightarrow$ set convention + HAC/block-bootstrapped ECM as the floor $\rightarrow$ account-level **balance-weighted, calibrated, competing-risk discrete hazard** with a latent common factor, validated **out-of-time first** and CPCV-second with embargo = H , aggregated to $S(t)$ with **time-block (not account-only) intervals** $\rightarrow$ deep/GenAI only if it earns

its keep on the same protocol → stress + validate with coverage as the headline. **Phases 0–3 already constitute a complete, defensible IRRBB deliverable; Phase 4 is the upside.**

10. Rigor cheat-sheet — what ported, what changed

Engine method	Run-off adaptation
CPCV primary (28 paths)	Walk-forward primary; CPCV secondary (forward extrapolation must be proven out-of-time). 8 super-blocks → 28 paths for variance only.
Embargo $E = \max$ horizon	Embargo/purge = H (run-off horizon backtested); 1-month for pure hazard.
Uniqueness sample weights	Balance weights ($CTRVL \ KDA$) + optional IPCW.
Cross-asset-honest block bootstrap (block = h)	Time-block bootstrap over months as the <i>headline</i> CI; account bootstrap is the small idiosyncratic part only.
Conformal (weighted/adaptive/CQR, Ch37)	Distribution-free coverage band on $S(t)$ /hazard residuals for stress; combined with block bootstrap; flagged as scenario-band (not certified) under OOD shocks.
SHAP / interpretability	3-tier explainability: ECM/logistic anchors → OOB SHAP + PDP/ALE (balance-weighted) → transformer attention + ablation (attention = hypothesis, not proof).
PIT calibration	Hazard reliability + PIT; calibration is the primary metric , not discrimination.
Step-3 global-fit look-ahead bug	<code>fit_end</code> required; buckets/scalers/ECM coeffs/frailty fold-local; macro PIT-aligned with publication lag.
IC-of-mean not mean-of- IC	Report aggregate $S(t)$ error + coverage , not the best-of-N fold; deflate HP/event-definition search.
Frozen test sanctity	Last 18–24 months frozen OOT, touched once; placement decided by Gate A macro inventory.
Latent factors / frailty	Latent monthly factor / account frailty so fan-charts aren’t under-dispersed.

11. Open decisions to confirm before coding

- Run-off quantity:** attrition-only, or attrition × balance-erosion $r_i(t)$? (§0, Phase 0)
- Event definition:** single event vs competing risks (closure/dormancy/floor)? (§3.1)
- Frozen-OOS length & placement:** default 24 months (2023–2024) — confirm after the Gate-A macro inventory (§1.1).
- Max evaluated horizon H :** sets the embargo and the limit of empirical (vs extrapolated) validation (§1.3, §1.1).
- Deep layer route if Gate B passes:** (A) per-account temporal model vs (B) foundation-model garnish (§7).